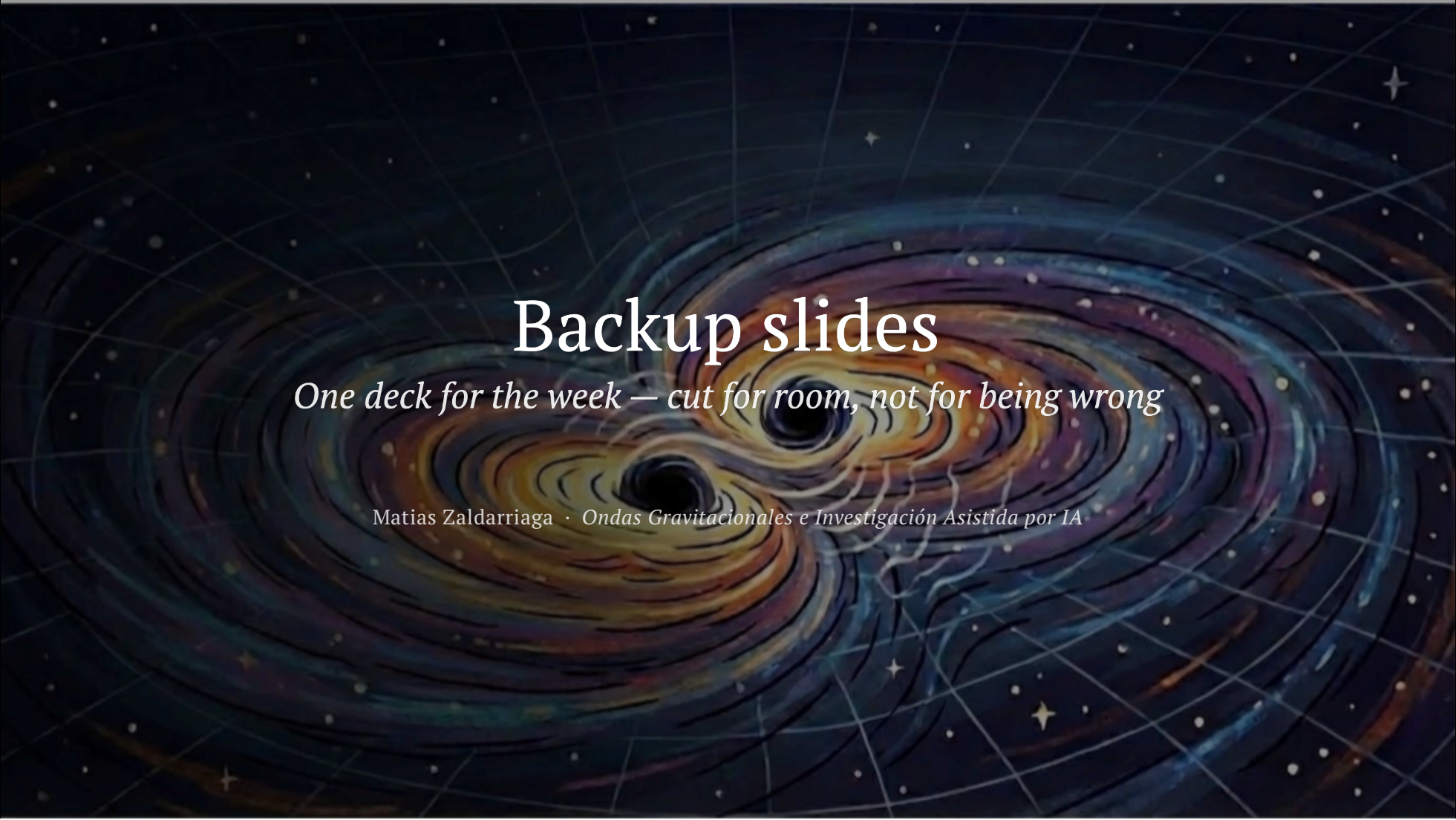


The background of the slide is a dark blue space filled with a grid of thin, light blue lines. Overlaid on this grid are two large, concentric, swirling patterns of light. These patterns are composed of many thin, curved lines in shades of blue, purple, and yellow, creating a sense of depth and movement. The overall effect is reminiscent of gravitational waves or a complex orbital system.

Backup slides

One deck for the week — cut for room, not for being wrong

Matias Zaldarriaga · Ondas Gravitacionales e Investigación Asistida por IA



DAY 1

Fundamentals

Peters, in *one slide*.

For a Keplerian binary of semi-major axis a and eccentricity e , averaged over one orbit:

$$\left\langle \frac{da}{dt} \right\rangle = -\frac{64}{5} \frac{G^3 m_1 m_2 (m_1 + m_2)}{c^5 a^3} \frac{1 + \frac{73}{24} e^2 + \frac{37}{96} e^4}{(1 - e^2)^{7/2}}$$

$$\left\langle \frac{de}{dt} \right\rangle = -\frac{304}{15} e \frac{G^3 m_1 m_2 (m_1 + m_2)}{c^5 a^4} \frac{1 + \frac{121}{304} e^2}{(1 - e^2)^{5/2}}$$

Circular case

$$t_{\text{merge}} = \frac{5}{256} \frac{c^5 a^4}{G^3 m_1 m_2 M}$$

The a^4 is why separation matters so violently.

Circularisation

Dividing the two gives $a(e)$ with the exponent $\frac{12}{19}$ and the famous $\frac{121}{304}$. Orbits circularise long before merger.

Eccentricity *enhancement*.

An eccentric orbit spends almost all its time out near apoapsis, moving slowly — but the emission goes as v^{10} , so the brief periastron passage dominates the radiated power and wins. At fixed semi-major axis the merger time collapses:

$$\frac{t_{\text{merge}}(e)}{t_{\text{merge}}(0)} \approx (1 - e^2)^{7/2} \quad \text{for } e \rightarrow 1$$

e	0	0.6	0.9	0.99
t_{merge} relative (exact Peters)	1	0.21	3.4×10^{-3}	1.7×10^{-6}

This is why dynamical formation channels — which produce eccentric binaries — can merge systems that circular decay would never bring in.

Hulse–Taylor has $e = 0.617$: its decay **rate** is enhanced by $F(e) = (1 + \frac{73}{24}e^2 + \frac{37}{96}e^4)/(1 - e^2)^{7/2} \approx 12$.

The *quantum* limit.

Two noise sources fight each other as you turn up the laser power:

Shot noise

Photon counting statistics. $\text{PSD}_{\text{shot}} \propto 1/(P_{\text{in}} G^2)$ — more power is better.

Radiation pressure

Fluctuating force on the mirrors. $\text{PSD}_{\text{RP}} \propto P_{\text{in}} G^2$ — more power is worse.

$$L^4 \text{PSD}_{\text{RP}} \text{PSD}_{\text{shot}} = \frac{\hbar^2}{M^2 (2\pi f)^4} \implies P_{\text{total}}^{\text{min}} = \frac{2\hbar}{M (2\pi f)^2}$$

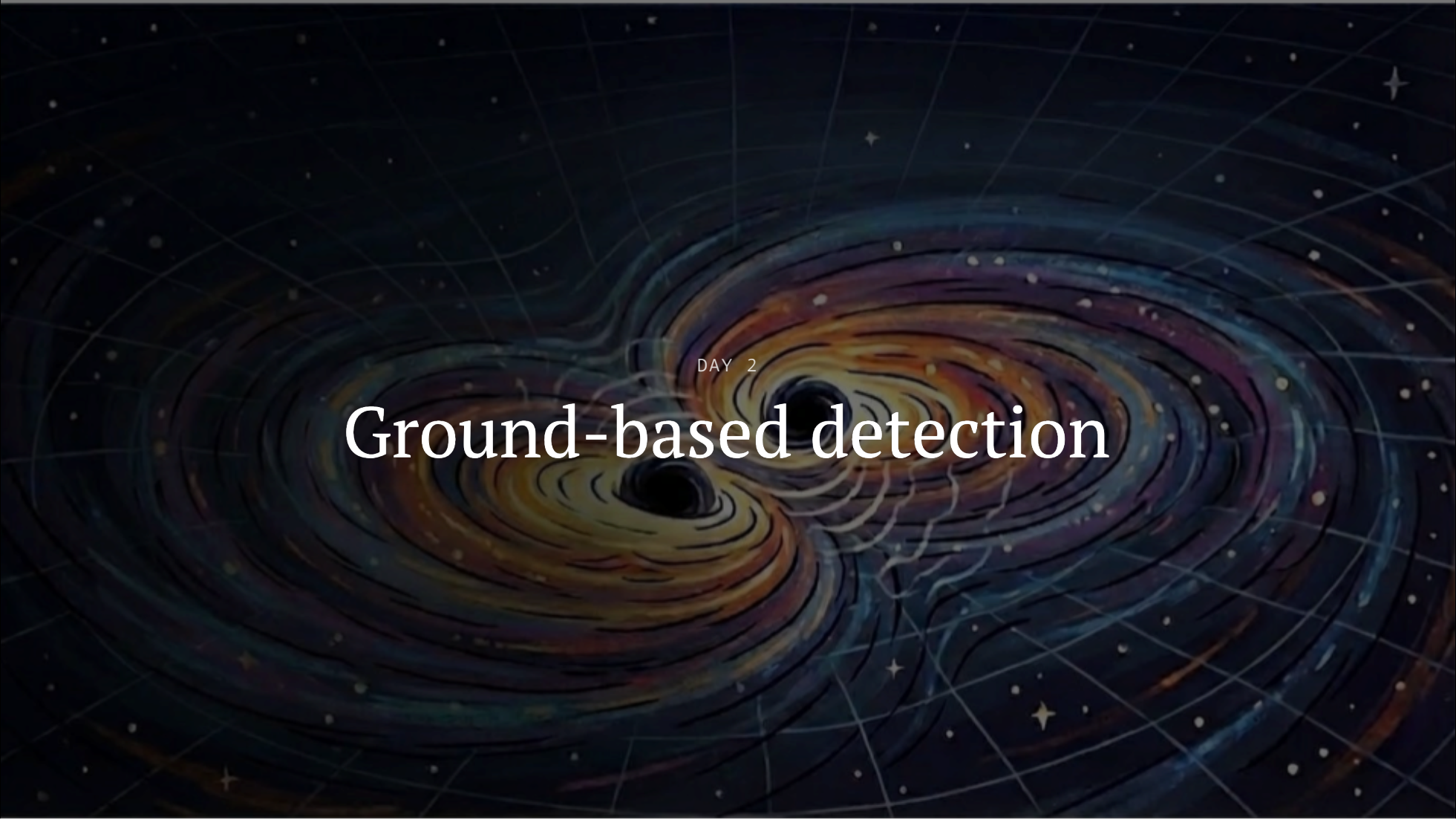
The product is independent of power — there is an optimum, and it is the *standard quantum limit*. Note it is not fundamental: it arises because we choose to measure **position** repeatedly, and the induced momentum uncertainty feeds back through the equations of motion. Measure something else and the limit changes. Squeezing beats it, and LIGO now uses squeezed light routinely.

Questions day 1 leaves open.

Two of day 1's reproductions have a question hung on them that we are not going to answer. Both are an evening's work, and the machinery is already on the machine in front of you.

- **the eccentric-decay question** — Peters gives the whole evolution with eccentricity and we used only the circular decay time. How much faster does a binary merge at high eccentricity, and which term in the evolution of e drives it?
- **the two source classes the week never returns to** — the landscape names four classes of source and this week follows two of them. Put a continuous wave and a stochastic background on the same axes, with the integration time each one needs.

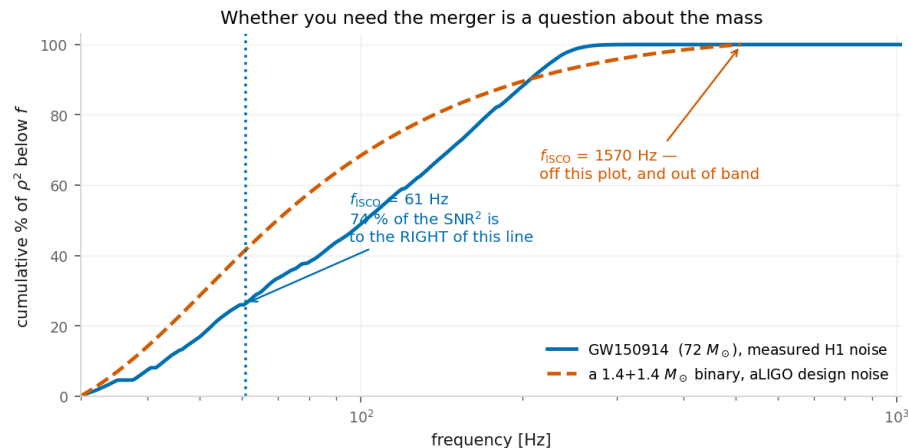
Each is an open question the course leaves you with. A row says what to start from and what an answer looks like, and that is all it says.

The background of the slide is a dark, deep blue space filled with a grid of thin, light blue lines representing a coordinate system. Two large, swirling, colorful structures resembling black holes or gravitational wells are the central focus. These structures are composed of concentric, wavy lines in shades of orange, yellow, and blue, suggesting intense energy and light. The two structures are positioned close together, appearing to be in the process of merging. In the space between and around these structures, there are several small, white, four-pointed star-like shapes, which likely represent gravitational waves or distant stars. The overall aesthetic is scientific and artistic, typical of modern astrophysical visualizations.

DAY 2

Ground-based detection

The inspiral *ends* where the band begins.



Cumulative SNR-squared with frequency; the inspiral ends where the band begins.

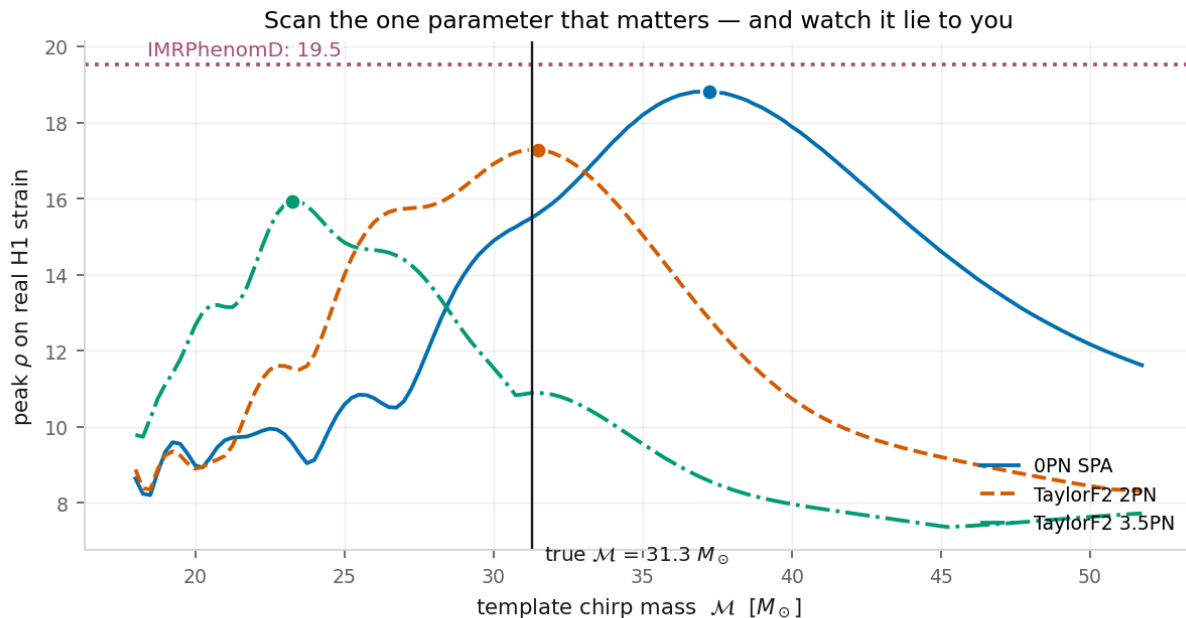
for GW150914 the inspiral ends where the band begins, so an inspiral-only template gives up half its SNR; for a neutron-star binary the inspiral is all there is to hear, which is why the same template is exact there and the difficulty moves to the number of cycles

For GW150914 the inspiral ends at $f_{\text{ISCO}} = 61$ Hz — where the band *begins*. **74 % of the signal-to-noise is above it.** Stop your template there and you throw away half your SNR; run it past there and you are filtering with a shape you cannot justify.

Which post-Newtonian order you kept barely matters: cut two of them at the same frequency and they differ by less than a tenth of a percent. What matters is where you stop, and no inspiral model tells you how to go further.

For a neutron-star binary f_{ISCO} is 1570 Hz. The merger is inaudible, and an inspiral template is not an approximation — it is the whole signal.

The chirp mass it gives you is *wrong*.



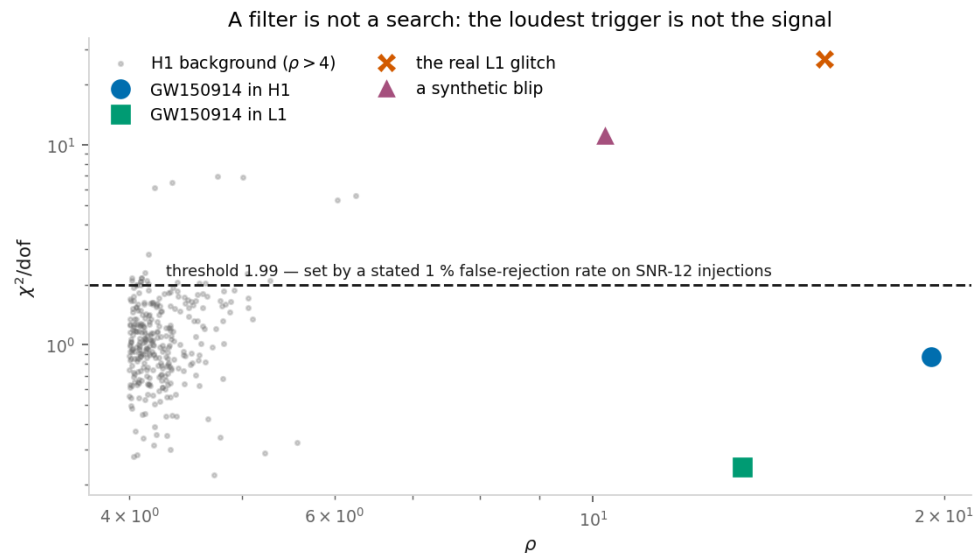
OPN peaks $+5.9 M_{\odot}$ from truth, 3.5PN -8.0 — the bias is a property of the template family, not a universal 'inspiral reads low'

OPN peaks **$5.9 M_{\odot}$ above** the truth; 3.5PN peaks **$8.0 M_{\odot}$ below** it. The inspiral-only template fakes missing merger power by pretending the binary is lighter; the OPN template, run past ISCO, does the opposite.

One fix, as an *example* — the χ^2 veto.

Ask whether the signal-to-noise arrived in the proportions the template predicts across the frequency band. A real signal does; a glitch dumps everything in one place.

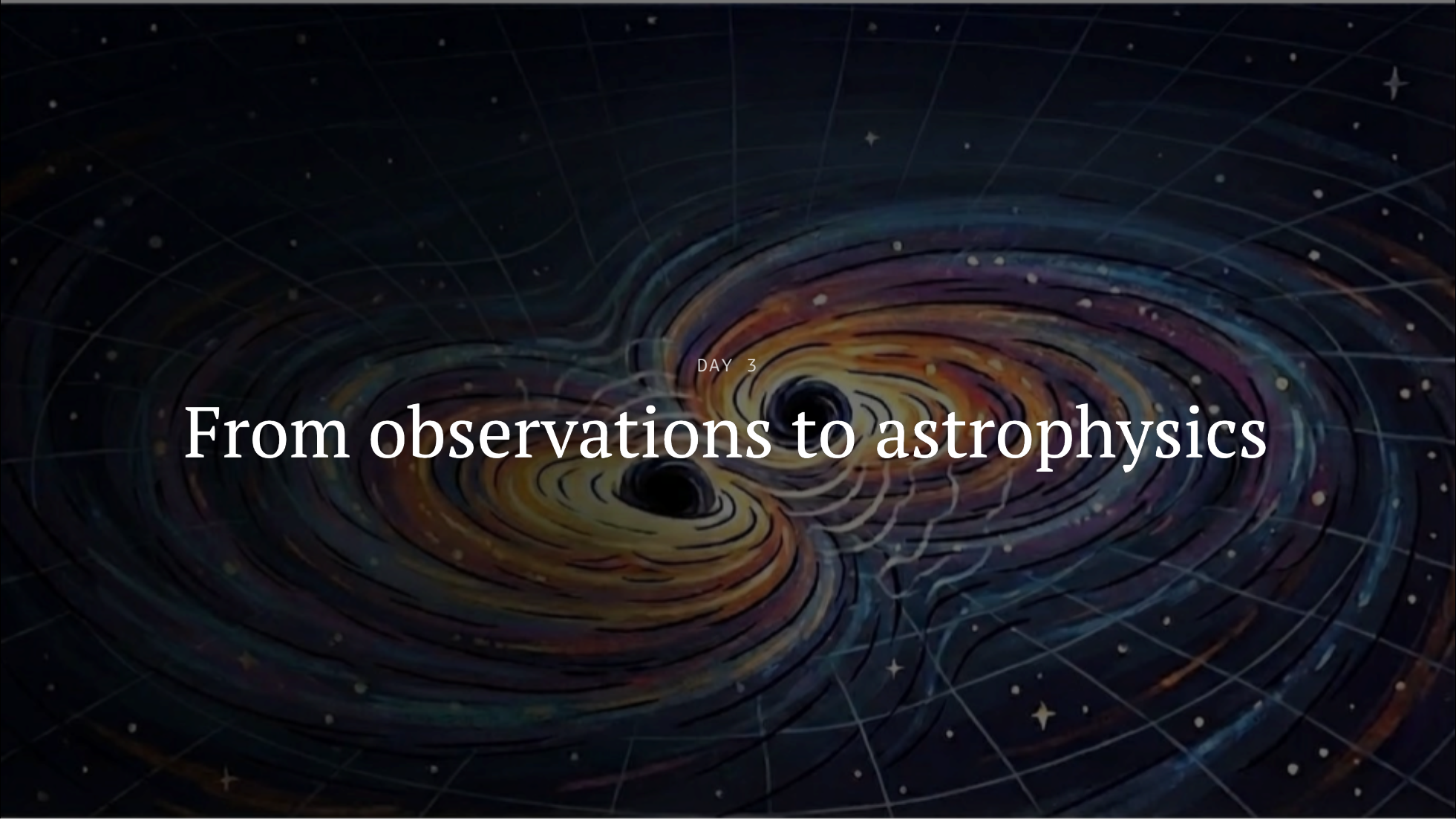
	ρ	χ^2/dof
signal-free background	—	1.00
GW150914, H1 / L1	19.5 / 13.4	0.87 / 0.24
the real L1 glitch	15.8	26.8



the veto removes 28/57 of the $\rho > 5$ background and the louder-than-the-event L1 glitch, and keeps both real triggers

Triggers in the SNR / reduced-chi-squared plane, with the calibrated threshold.

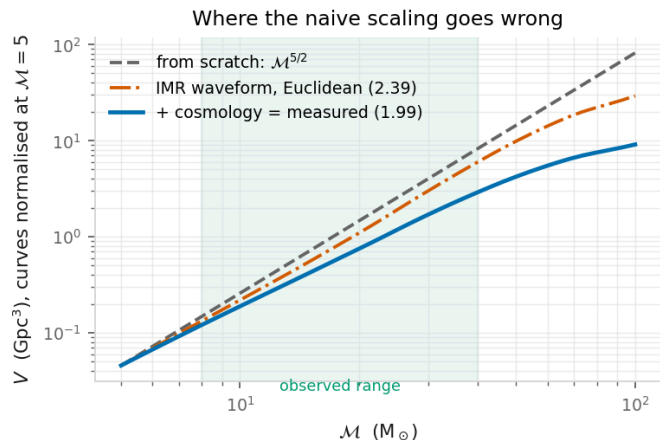
One example from a whole toolbox — data-quality flags, coherence tests, time slides (shifting one detector's clock to manufacture signal-free background). The sense to keep is only this: **real data is more complicated than the Gaussian story**, and every fix is calibrated against a stated error rate, not by taste.

The background of the slide is a dark, deep blue space filled with a fine grid of light blue lines, resembling a celestial map or a coordinate system. Two large, spiral galaxies are depicted in the process of merging. The galaxies have vibrant, multi-colored spiral arms in shades of orange, yellow, and blue. The central regions of the galaxies are dark, suggesting the presence of supermassive black holes. The overall effect is a dramatic and artistic representation of cosmic events.

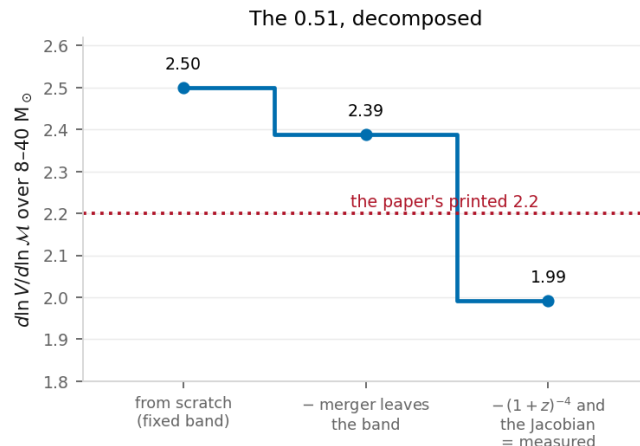
DAY 3

From observations to astrophysics

Where the 0.5 went



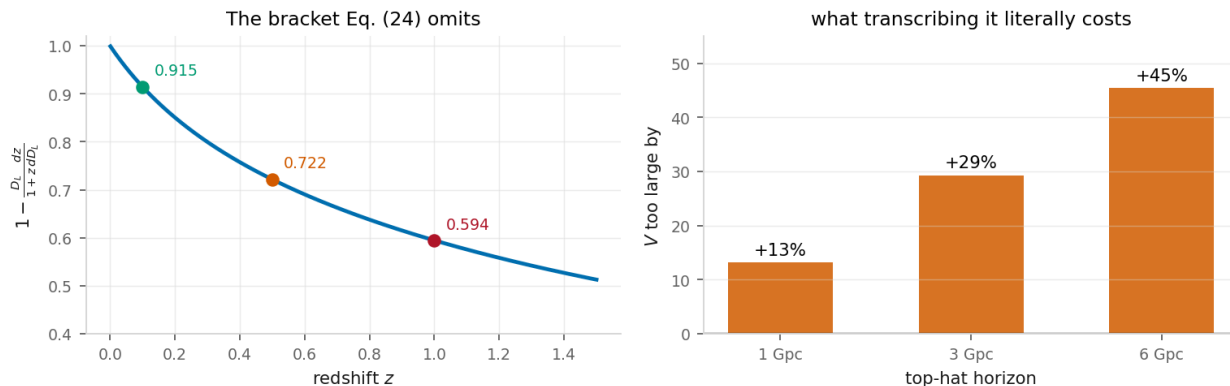
Derive 2.5, measure 1.99, and the plan's stated cause turns out to be the smaller half of it. Inside the observed range the cosmological factors cost 0.40 of slope and the merger leaving the band costs 0.11; the band argument only takes over above 60 M_{sun} , where the slope falls to 0.93. We get 1.99 against the paper's printed 2.2 and record that as a disagreement a choice could explain -- the band, the approximant and the PSD shape are all unstated in the paper -- rather than tuning toward it. NONE of this is in the gate: see the next figure for why.



The sensitive-volume slope: derived, measured, and decomposed.

Derive 2.5, measure 2.0, decompose the difference. Inside the observed range the cosmological factors cost **0.40** of slope and the merger walking out of the band costs **0.11** — the band argument only takes over above 60 $M_{\odot}$, where the slope falls to 0.93.

The erratum this calculation carries



1806.10610 Eqs. (13) and (24) drop the Jacobian bracket of the comoving-to-luminosity change of variable, and THE GROUP'S OWN SUCCESSOR PAPER reinstates it: 2008.07014 Eq. (30) carries it and derives it as $(dt_c/dt)(dV_c/dV_L)$, four years later. Transcribed literally it makes V too large by 13, 29 and 45 per cent out to 1, 3 and 6 Gpc. It is transcribed wrongly exactly once, here, so that the size of it is a number; everywhere else in this exercise it is correct. Note what it is NOT: it sits inside G1's factor-of-two tolerance, so this gate cannot catch it, and saying that out loud is more honest than pretending a gate catches everything.

The dropped Jacobian, and what it costs with distance.

Eqs. (13) and (24) drop the Jacobian of the comoving-to-luminosity change of variable. **The group's own successor paper reinstates it** — Roulet et al. 2020 Eq. (30) carries the bracket and derives it, four years later.

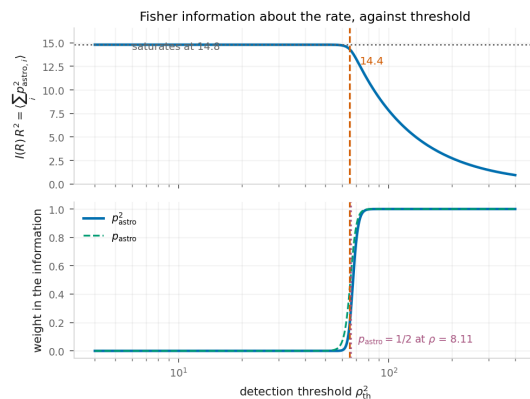
That is a story about how correction actually happens, not a gotcha.

The information in a *trigger*.

$$I(R) = \frac{1}{R^2} \left\langle \sum_i p_{\text{astro},i}^2 \right\rangle$$

Each trigger carries information with weight p_{astro}^2 .

One expression, and it justifies **both** including marginal events **and** not going arbitrarily deep.



2008.07014 Fig. 3, from a closed form in twenty lines and with no download at all. A power-law foreground in a Gaussian background, normalised to the O1+O2 numbers $N_{\text{a}}(\rho_{\text{th}}^2 > 65) \approx 15$ and $N_{\text{b}}(\rho_{\text{th}}^2 > 65) = 1$. Eq. (22) says each trigger enters the Fisher information with weight p_{astro}^2 , so lowering the threshold from $\rho_{\text{th}}^2 \approx 65$ to 4 admits vastly more triggers and buys 3 per cent more information. One picture that justifies both including marginal events and not going arbitrarily deep – and the reason the IAS catalogue's extra events were worth adding: they raised the summed p_{astro}^2 by 50 per cent, shrinking uncertainties by 0.82.

The information in a trigger, weighted by p -astro squared.

Both integrals are somebody else's samples

The numerator is an average over **posterior samples**. The denominator is a sum over **found injections**. Neither is an integral you did.

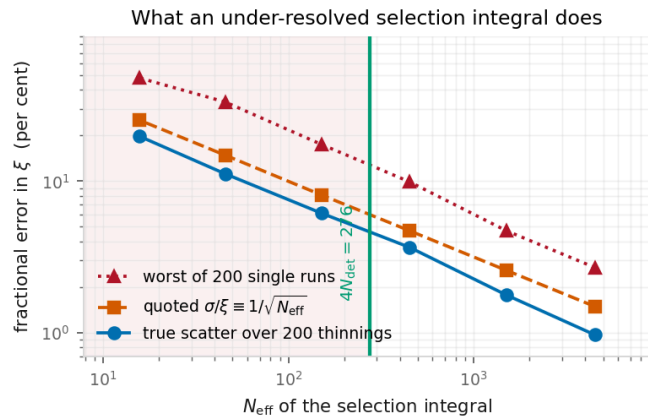
$$N_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}, \quad \text{require } N_{\text{eff}} > 4N_{\text{det}}$$

And there is an identity worth seeing once:

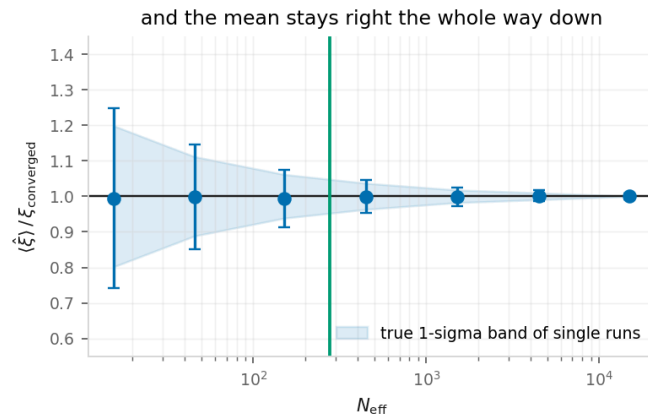
$$\frac{\hat{\sigma}}{\hat{\xi}} = \frac{\sqrt{\sum w^2}}{\sum w} = \frac{1}{\sqrt{N_{\text{eff}}}}$$

Quoting an error bar and quoting N_{eff} are **the same statement**. The difference is that N_{eff} gets compared against $4N_{\text{det}}$, and a σ gets compared against nothing.

The estimator, *starved*.



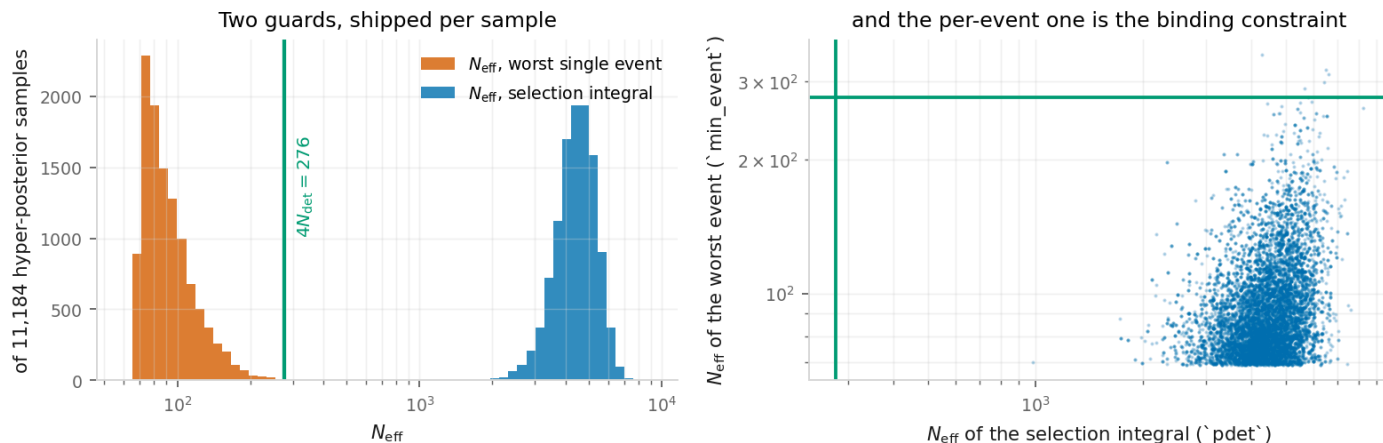
The estimator is **UNBIASED** — the mean of 200 independent thinnings is within 0.5 per cent of the converged answer even with 36 found injections left. So 'is the answer biased' is the wrong question. What explodes is the **SCATTER**: single runs land up to 48 per cent away, and there is no way to tell from one run which one you got. The run's own error bar is honest — it tracks the true scatter to a constant 0.74x — and it does not save you, because $\sigma/\xi = 1/\sqrt{N_{\text{eff}}}$ IDENTICALLY, so quoting an error bar and quoting N_{eff} are the same statement. The difference is that N_{eff} gets compared against $4N_{\text{det}}$ and a sigma gets compared against nothing. N_{eff} crosses that line while the true scatter is still 6 per cent.



Thinning the injection set: unbiased mean, exploding scatter.

The estimator is **unbiased** all the way down — the mean of 200 thinnings is within 0.5 % with 36 injections left. What explodes is the **scatter**: single runs land up to 48 % away, and nothing in one run tells you which you got.

The collaboration ships its own guard

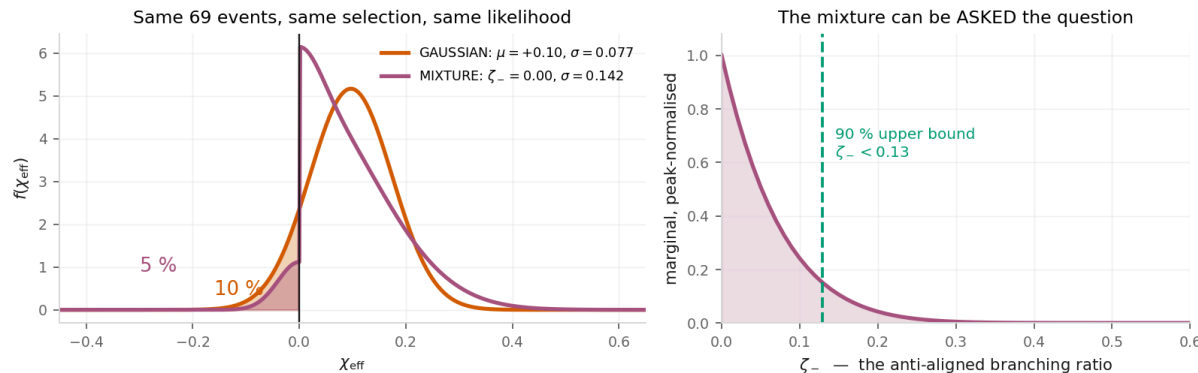


The GWTC-3 population release ships ``min_event_n_effective`` and ``pdet_n_effective`` for every one of its 11,184 hyper-posterior samples. The collaboration's own convergence criterion is a **COLUMN**, not a citation, and that is the argument for building day 3 on this release. Note which one binds: the selection integral sits at a median of 4458 and clears 4 N_{det} on 100 per cent of samples, while the worst single event sits at 86 and clears it on 0.1 per cent. The guard is not a formality that always passes -- which is exactly why it is worth shipping.

The shipped convergence columns, per hyper-posterior sample.

`min_event_n_effective` and `pdet_n_effective` , **per hyper-posterior sample**, 11,184 of them. The convergence criterion is a **column**, not a citation — and the per-event guard is the binding one.

Spins — and now the answer depends on the model



Two parametrisations, one dataset. Under the GAUSSIAN, 'how much is negative' is not a free parameter -- it is fixed by μ/σ , so the width and the negative fraction have ONE answer between them, and here that is 10 per cent. The mixture has ζ_- as an explicit parameter and can be asked directly: it says zero, with a 90 per cent bound of 0.13. DISAGREEMENT, documented: the LVK's own GAUSSIAN fit publishes 29 (+15/-13) per cent where ours gives 10; theirs is a joint χ^2_{eff} -and- χ^2_p fit over full posteriors with the full selection function and ours is χ^2_{eff} -only at one mass. The qualitative claim survives and the magnitude does not reproduce, and that is reported rather than smoothed over. Day 3 does not adjudicate which model is right.

The same events under two spin parametrisations.

Same 69 events, same selection function, same likelihood. Under the GAUSSIAN, "how much is negative" is **not a free parameter** — it is fixed by μ/σ . The mixture has ζ_- explicitly and can be **asked**: it says zero, bound < 0.13 .

The same width, two *readings*.

LVK's GAUSSIAN model publishes 29^{+15}_{-13} % of binaries at $\chi_{\text{eff}} < 0$. Roulet et al. 2021 gets $\zeta_- \approx 0$ from the same data — and reproduces the LVK number exactly when it adopts the LVK model.

Both agree on the width ($\sigma_{\chi_{\text{eff}}} \approx 0.11$ implied against $0.13^{+0.12}_{-0.07}$) **and disagree on what it means.**

That is a sharper framing than "they disagree about negative spins". We are not adjudicating it. Attribute it and move on.

Our own version gets 10 % rather than 29 % — a χ_{eff} -only fit at one mass against their joint fit over full posteriors. The qualitative claim survives, the magnitude does not, and the exercise says so.

If it comes down to one item, make it this one

GW170817A's p_{astro} moved from **0.07** to **0.22–0.26**

purely because the favoured **mass model** changed.

A number that is supposed to answer "*is this real*" turns out to depend on the population you assumed — which is the day's own lesson turned on the day's own machinery.

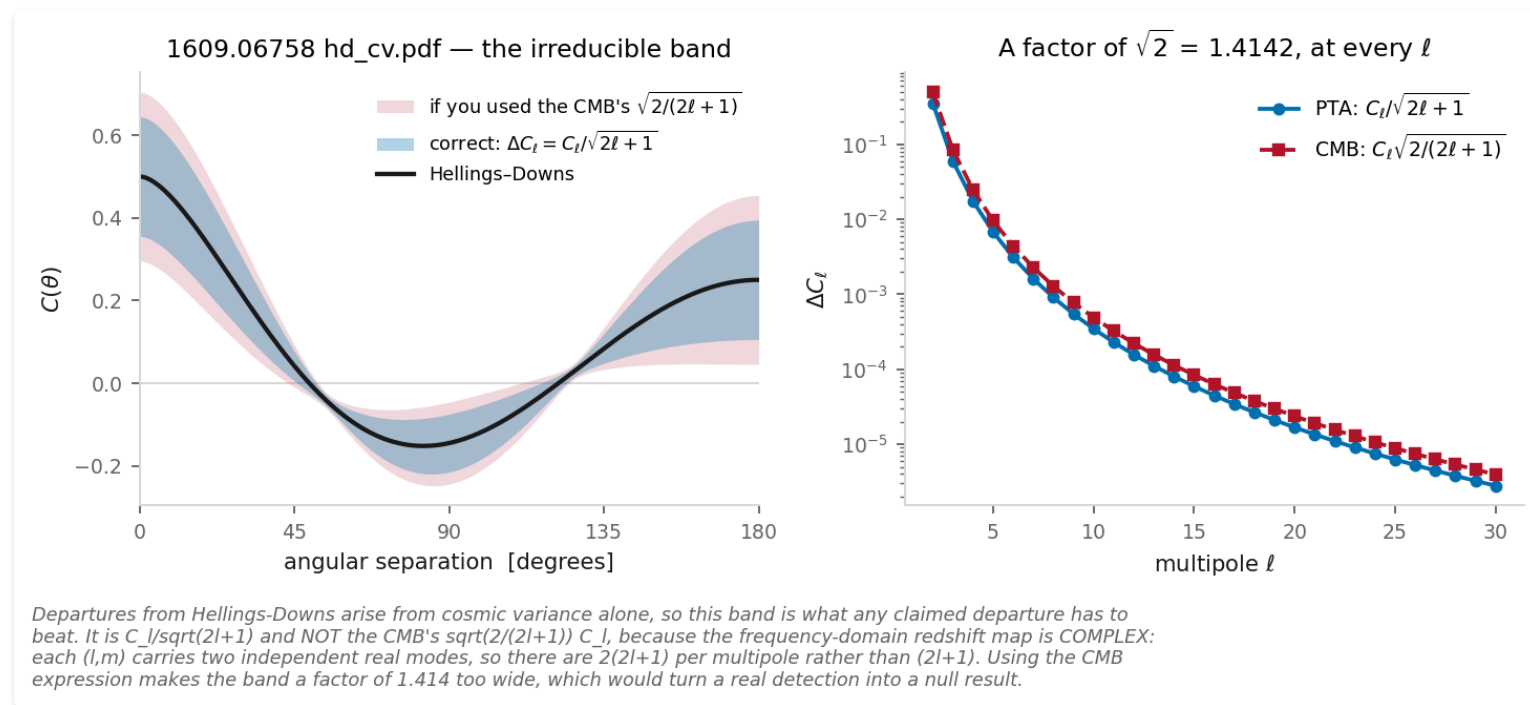
Thirty seconds, and it needs no position taken.

The background of the slide is a dark, deep blue space filled with a grid of thin, light blue lines representing a coordinate system. Two large, swirling, colorful structures resembling black holes or gravitational wells are the central focus. These structures are composed of concentric, wavy lines in shades of orange, yellow, and blue, suggesting intense energy and light. The two structures are positioned close together, appearing to be in the process of merging. Scattered throughout the background are numerous small, white, four-pointed stars of varying sizes. The overall aesthetic is artistic and scientific, evoking the theme of astrophysics.

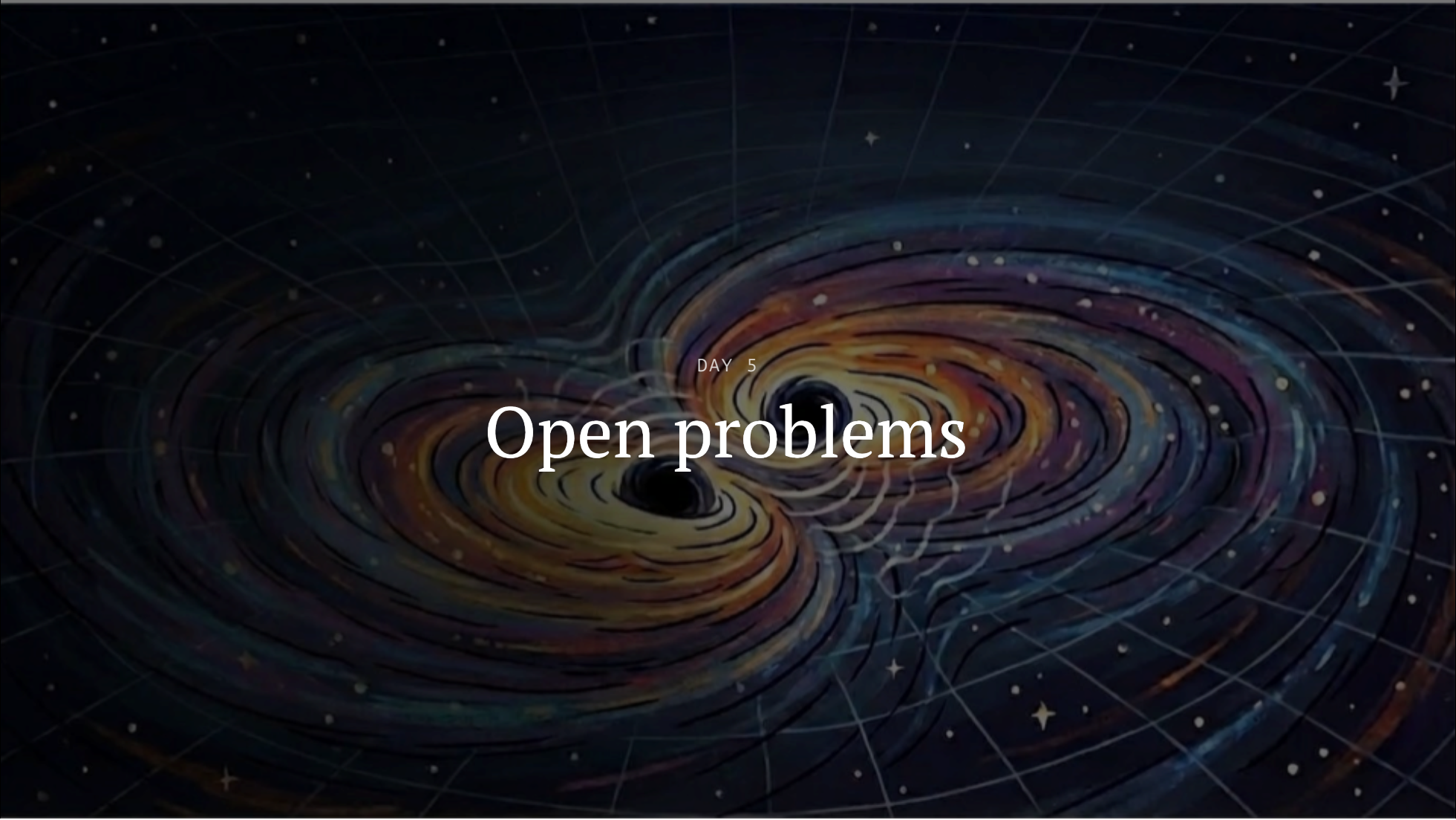
DAY 4

Pulsar timing arrays

What a departure would have to *beat*.



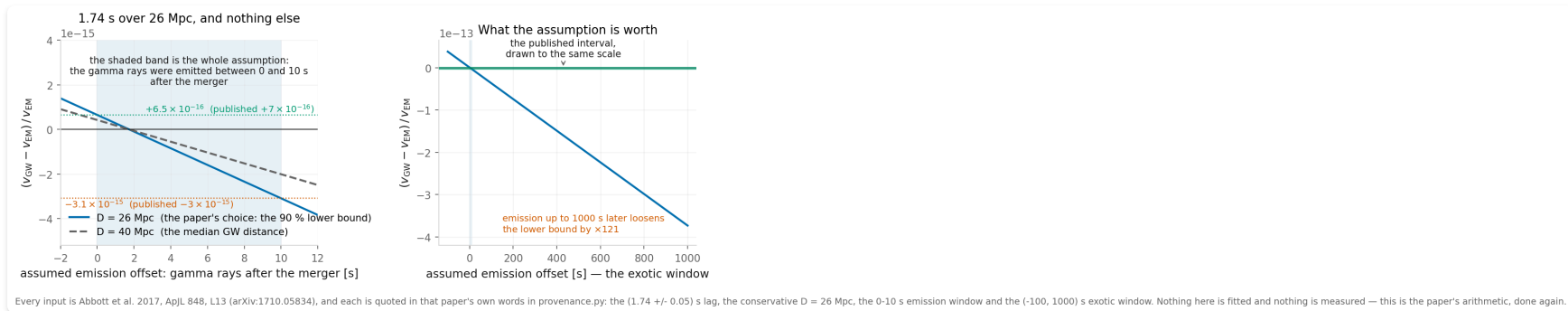
The band is $C_\ell/\sqrt{2\ell+1}$, **not** the CMB's $\sqrt{2/(2\ell+1)}$ — the frequency-domain map is complex and carries twice the modes.

The background of the slide is a dark, deep blue space filled with numerous small, white, four-pointed stars. A faint, light blue grid of lines is overlaid on the entire scene, creating a perspective of depth. In the center, two large, spiral galaxies are in the process of merging. Their swirling arms, rendered in shades of orange, yellow, and brown, are being pulled together by gravity, creating a complex, tangled pattern of light and dark matter. The galaxies' cores are dark and dense, with bright, glowing regions at their centers. The overall effect is one of cosmic drama and the dynamic nature of the universe.

DAY 5

Open problems

The speed of *propagation*, as a picture.



Left: the two published limits, reproduced from the lag, the distance and the emission window. Right: the paper's own caveat — exotic scenarios widen the window to $(-100, 1000)$ s — drawn to the same scale.